Probabilistic Testing for NLP Systems

A Reproducible Approach to Evaluating Stochastic Behaviour in Real-World Applications

Abstract

Natural language processing systems are increasingly deployed in settings where output quality matters operationally, yet the systems themselves are often stochastic. This is particularly true for LLM-based applications, but the problem also arises more broadly wherever NLP pipelines depend on probabilistic models, retrieval components, ranking, sampling, or other non-deterministic mechanisms.

In such contexts, traditional pass/fail testing and one-shot benchmark scores are often insufficient: they obscure variance, conceal instability, and provide limited guidance when a system appears to “sometimes work”.

This talk presents a probabilistic testing approach for NLP systems, embodied by the PUnit framework. Rather than treating a single execution as decisive, the approach evaluates behaviour over repeated trials. For binary success/failure criteria, individual outcomes can be modelled as Bernoulli variables and summarised by the observed success rate. This makes it possible to assess whether an NLP system still behaves within an expected range, whether a drop in quality/accuracy is likely to have occurred (following a change of model, training data update etc.), and whether differences between versions are meaningful or merely noise.

The PUnit approach is relevant to a wide spectrum of NLP applications, including text classification, information extraction, retrieval-augmented generation, summarisation, translation, and agentic workflows built on top of language models.

The emphasis is not on replacing existing benchmarks, but on complementing them with an evaluation layer that is more suitable for production-facing, stochastic systems. In particular, the method supports a more reproducible and transparent interpretation of quality by making uncertainty, sample size, and tolerance explicit.

The talk will outline the underlying testing model, show how probabilistic verdicts can be integrated into engineering workflows, and discuss how this style of evaluation can help bridge the gap between academic NLP metrics and the realities of deployed systems.

The talk will incorporate a live demonstration of the PUnit probabilistic testing framework.

CV

Mike Mannion is a is a long-time software engineering professional at Karakun AG in Basel and leads Karakun's AI working group. He has worked in a wide range of industries, but always with a special interest in software quality and the application of statistics in quality assurance and testing. In recent years he has turned his attention to AI applications, testing AI applications, and the wider challenge of testing non-deterministic systems. He is the creator of the probabilistic testing framework PUnit, which is helping companies get to grips with testing services reliant on LLMs. Mike holds a bachelor's degree in computer science from the University of Brunel (UK) and a dual MBA from the universities of Bern (CH) and Rochester (NY, USA), and a Professional Certificate in AI and Machine Learning from Imperial Collage London.

Intended audience

Researchers and practitioners interested in NLP evaluation, benchmarking, testing, LLM quality assurance, and production reliability.